

Question Bank: Object Oriented Programming with JAVA (BCS306A)

1. Explain the concept of multidimensional arrays in Java. Write a program to print the elements of a 2D array using nested loops.
2. Explain 2D Arrays with syntax and an implementation example.
3. Describe how memory is allocated for a 2D array in Java with a suitable example program.
4. Write a program illustrating the difference between if-else and else-if statements.
5. Describe object-oriented concepts, including their benefits and how it can be implemented.
6. Describe OOP methodology and demonstrate how Java supports each OOP principle with examples.
7. Write a Java program to calculate the factorial of a given number using a recursive method.
8. Write a Java program to perform matrix multiplication using 2D arrays.
9. Explain the advantages and limitations of recursion.
10. Write a Java program to calculate the sum of two matrices.
11. Illustrate Overriding Methods with an example.
12. Explain the use of the super keyword in inheritance with examples.
13. Explain different types of if statements in JAVA.
14. Explain runtime polymorphism in Java with a suitable method overriding example.
15. Explain the advantages and limitations of recursion. Write a Java program to calculate the factorial of a given number using a recursive method.
16. Write a Java program to demonstrate dynamic method dispatch using a base class Shape and derived classes Triangle, Circle and Rectangle, overriding a calculateArea() method.
17. Illustrate constructors and constructor overloading with examples.
18. What are constructors? Give the types and explain the properties of constructors. Support with appropriate examples.
19. Show how overriding works between a parent class Animal and child classes Dog and Cat using a sound() method.
20. With a java program, illustrate the use of the ternary operator to find the greatest of three numbers.